TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

1. What is the solution set of $-4x > 12$?

A. $x > -3$

B. $x < 3$

C. $x > 3$

D. $\boldsymbol{x < -3}$

Divide by -4 and flip: $x < -3$.

2. What is the solution set of $2x + 3 \leq x - 1$?

F. $\boldsymbol{x \leq -4}$

G. $x \geq -4$

H. $x \leq -2$

J. $x \leq 4$

Subtract x: $x + 3 \leq -1$. Subtract 3: $x \leq -4$.

3. Which inequality means the same as $-6 \leq x$?

A. $x \leq -6$

B. $\boldsymbol{x \geq -6}$

C. $x > -6$

D. $x < 6$

Read it from x's side: x is greater than or equal to -6.

4. What value of x makes $3(x - 2) = 9$ true?

F. 3

G. 11/3

H. **5**

J. 1

$3x - 6 = 9$, $3x = 15$, $x = 5$.